



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
An Introduction To Bayesian Analysis (MTH535A)
End Semester Examination
Date: April 27, 2023, Thursday

Timing: 8:00–9:30 AM (8:00–10:00 AM for DAP) Spring '23 Max point: 20(+5 bonus)

1. “The support of a posterior distribution is always a subset of the support of the prior distribution”-Justify mathematically. (2 points)
2. Let $Y|\mu \sim \text{Normal}(\mu, 1)$ and μ has an improper prior $\pi(\mu) = 1, \mu \in \mathbb{R}$. Show that the posterior distribution of μ is a proper PDF. (2 points)
3. Let $Y|\mu \sim 0.5\text{Normal}(\mu, 1) + 0.5\text{Normal}(0, 1)$ and μ has an improper prior $\pi(\mu) = 1, \mu \in \mathbb{R}$. Show that the posterior distribution of μ is also improper. (2 points)
4. We want to decide between two models

$$\mathcal{M}_1 : Y \sim \text{Poisson}(\lambda_0), \quad \mathcal{M}_2 : Y|\lambda \sim \text{Poisson}(\lambda), \quad \lambda \sim \text{Exponential}(\lambda_0^{-1}).$$

Find the Bayes factor $BF_{2,1}$ in terms of λ_0 and Y . (2 points)

5. Consider a four-way random effects model $Y_{ijklm} = \alpha_i + \beta_j + \gamma_k + \delta_l + \varepsilon_{ijklm}$, where $\alpha_i \stackrel{IID}{\sim} \text{Bernoulli}(p_\alpha)$, $\beta_j \stackrel{IID}{\sim} \text{Bernoulli}(p_\beta)$, $\gamma_k \stackrel{IID}{\sim} \text{Bernoulli}(p_\gamma)$, $\delta_l \stackrel{IID}{\sim} \text{Bernoulli}(p_\delta)$, and $\varepsilon_{ijklm} \stackrel{IID}{\sim} \text{Bernoulli}(p_\epsilon)$. Find the correlation between $Y_{i_1 j k_1 l m}$ and $Y_{i_2 j k_2 l m}$ where $i_1 \neq i_2$ and $k_1 \neq k_2$. (2 points)
6. Suppose we have a multiple linear regression setting $Y_i \stackrel{Indep}{\sim} \text{Normal}(\mathbf{X}_i' \boldsymbol{\beta}, \sigma^2), i = 1, \dots, n$. For some i 's, the $\mathbf{X}_i$ are fully missing (other $\mathbf{X}_i$'s are not partially missing), and we want to impute them by assuming $\mathbf{X}_i \stackrel{IID}{\sim} \text{Normal}_P(\boldsymbol{\mu}_X, \sigma_X^2 \mathbf{I}_P)$. Choose noninformative conjugate priors for unknown parameters and hyperparameters. Write down a Gibbs sampling algorithm along with detailing each full conditional posterior distribution for all the necessary terms. (6 points)
7. Under the loss $\ell(\hat{\theta}, \theta) = |2\hat{\theta} - \theta|$, find the Bayes rule. For a discrete parameter space, show that the MAP estimator is the Bayes rule under a 0-1 loss function. (2+2 points)
8. The following is a R snippet. Find the unconditional joint distribution of $\mathbf{X}$. (2 points)

```
theta <- rnorm(n = 1, mean = 2, sd = 0.5)
X <- rnorm(n = 4, mean = theta, sd = 1)
```

9. The following is a R snippet. Find the unconditional joint distribution of $\mathbf{X}$. (2 points)

```
theta <- rnorm(n = 2, mean = 2, sd = 0.5)
X <- rnorm(n = 4, mean = theta, sd = 1)
```

10. The following is a R snippet. Find the unconditional joint distribution of $\mathbf{X}$. (1 point)

```
theta <- rnorm(n = 4, mean = 2, sd = 0.5)
X <- rnorm(n = 4, mean = theta, sd = 1)
```